

Global Environmental Politics

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Last class: climate change and governance

- The regime complex for climate change can be inefficient.
- Makes it hard to standardize and generate effective cooperation.
- Competition generates forum shopping and race to the bottom.
- Can generate poorly designed agreements.
- Requires vertical coordination.
- Feasibility and incentive compatibility are necessary.
 - Feasibility: is it possible under current constraints.
 - Incentive compatible: No opposition + tying hands.

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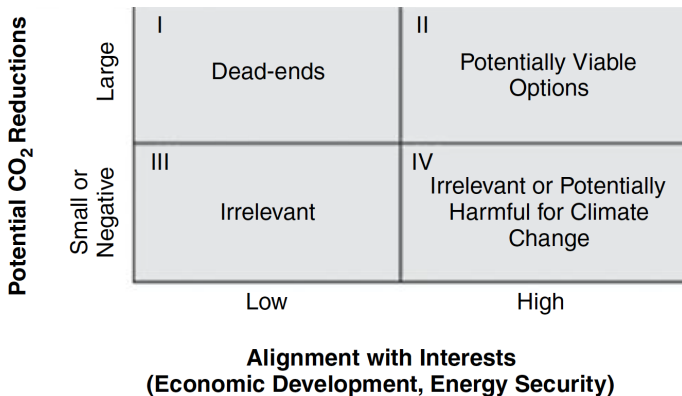
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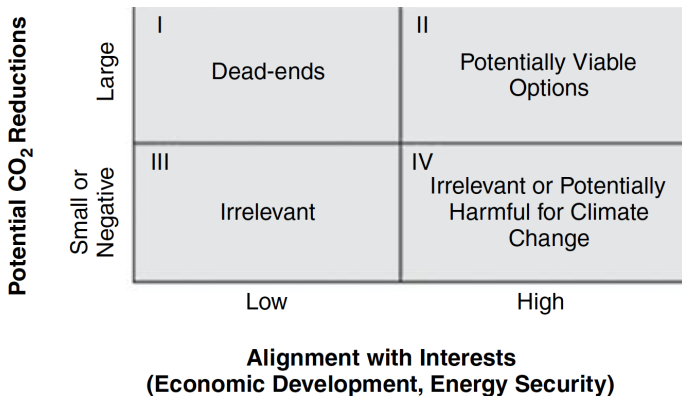
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Potential outcomes of climate change agreements



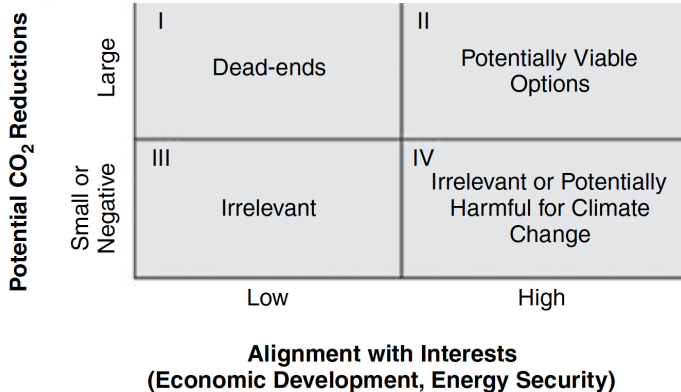
- (I): No agreement.
- (II): Positive agreements (redistributive and higher well-being).
- (III): Poorly designed agreements.
- (IV): Negative agreements (redistributive and lower well-being).

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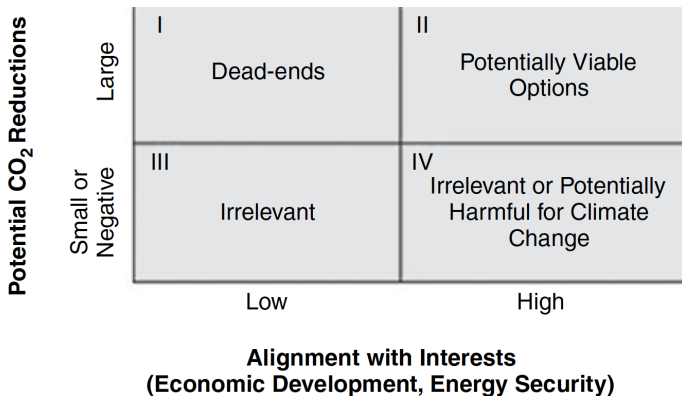
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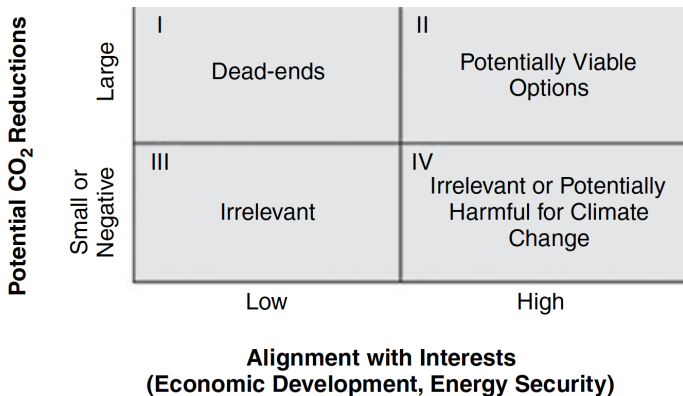
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(II) Positive agreements



- (IIa) The development model (e.g., IOs).
- (IIb) Politically cheap but limited impact.
- (IIc) Situational constraints; depend on innovation reducing cost.
- (IId) Feasible but v. difficult (high capacity, low corruption).

(II) Positive agreements

Administrative Feasibility	High	<u>Outsiders Help with Technical & Financial Support</u> Classic Offsets Projects IIa	<u>Co-benefits Situations</u> False Offsets Projects IIb
	Low	<u>Multi-Prong Assistance Projects</u> Requires administrative, financial and technical assistance IIc	<u>Administrative Capacity-building</u> IId
		Low	High
		Practical Ability (Finances, Technology)	

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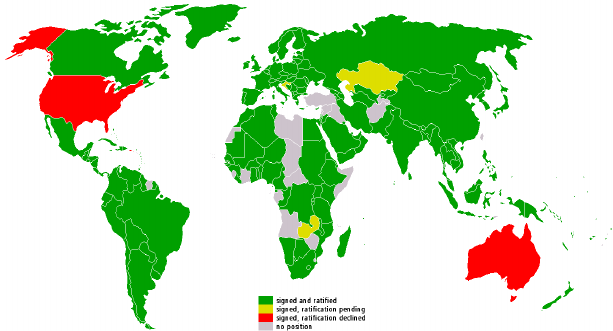
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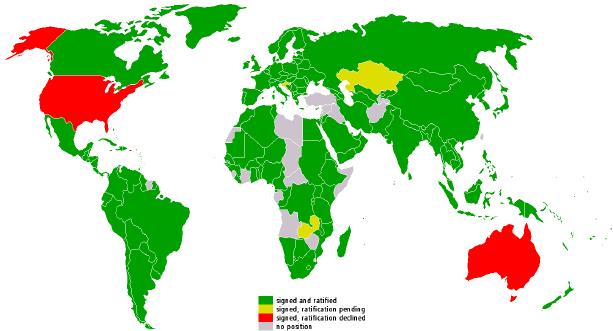
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Limitations for achieving II: Hegemons and veto players



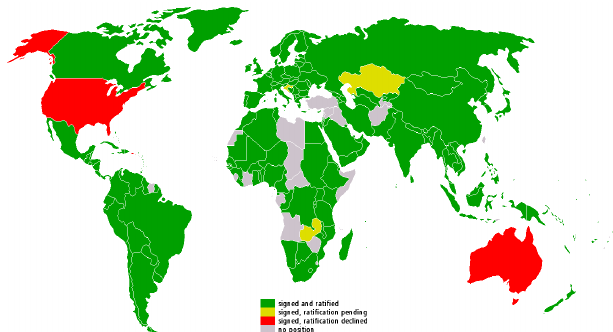
- Hegemons and other big players may be necessary.
- Support from a winning coalition is necessary (size?).
- Emitters resisted Paris agreement commitments.

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Table 1 Likely orientations of intergovernmental organization members as a function of power configurations

		Power in the basic game	
		Weak	Strong
Power in the negotiation game	Strong	Prefers strong organization	Prefers moderately strong organization
		Contributes more than expected given its capabilities	Contributes at about the level expected given its capabilities
	Weak	Sees for itself few, if any, benefits from membership	Prefers 1. Organizational reform 2. A weak organization
		Passive member, paying at most a small contribution to organizational activities	Contributes less than expected given its capabilities; may well work through other channels or at least explore alternative arrangements

- Uncertain cost/benefits for hegemons and veto players.
- Trade-off between depth and participation.
- Domestic support is important for generating int' cooperation.

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- Domestic support is important for generating int' cooperation.
- Cities, firms, voters, etc. which can also act as veto players.

Feasible alternatives

- Independent information collection (investigations) by IO can alleviate non compliance.
- If IO has enforcement power, $\Downarrow$ need of investigative resources.
- Deviation implies punishment via sanctions.
 - Effectiveness depends on economic interdependence.
- Level of compliance may depend on alignment with hegemon.
 - Higher if hegemon complies; lower if he doesn't.

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The game of international climate agreements

- Players or actors involved.
- Rules of the game: access, procedural and decision rules.
 - Who can do what (i.e., actions).
 - Who knows what about the state of things and interests (i.e., information).
 - What is the distribution of interests and power.
 - Who moves or decides first, who follows, and so on...
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Deadlock: making the basic game work

Note there is no Nash in this game:

		B	
		Cooperate	No cooperate
A	Cooperate	1.5, 1	-2, 1.5
	No cooperate	1, 0	0, -1

How can we create a Nash equilibrium?

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Note *{cooperate, cooperate}* can be a Nash.

Information: making the basic game work

A plays cooperate with $q \in [0, 1]$ and B plays cooperate with $p \in [0, 1]$

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If side payment:

$$p - 2(1 - p) \geq p + 0(1 - p) \quad \text{and} \quad 1.5q + 0(1 - q) \geq 1.5q - (1 - q)$$

$$3p - 2 \geq p \quad \text{and} \quad 1.5q \geq 2.5q - 1$$

$$p \geq 1 \quad \text{and} \quad 1 \geq q$$

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 2.5p - 2 &\geq p \quad \text{and} \quad q \geq -1 + 2.5q \\
 p &\geq 4/5 \quad \text{and} \quad 2/3 \geq q
 \end{aligned}$$

Compliance: hegemons and punishment

Play the game repeatedly:

		B	
		Cooperate	No cooperate
A	Cooperate	1, 1.5	-2, 1.5
	No cooperate	1, 0	0, -1

Assume $p(n) \in [0, 1]$ falls with group size (n):

$$\begin{aligned}
 & \sum_{t=0} \delta^t [p(n-1) - 2(1 - p(n-1))] > \\
 & ([1 \cdot p(n-1) - 0(1 - p(n-1))] + \sum_{t=1} \delta^t [0 + p(n-1)/t]; \\
 & \frac{3p(n-1) - 2}{1 - \delta} > p(n-1); \\
 & \delta > 2 \left(\frac{1 - p(n-1)}{p(n-1)} \right).
 \end{aligned}$$

Incentive compatible agreements and compliance

- Proper pay-offs can encourage participation (basic game).
- ...then we need to secure compliance (power game).
 - A ruling coalition may be sufficient.
 - ...but needs to be incentive compatible.
 - Too stringent and few will join.
 - Too flexible and few will comply.
- May need participation of compliance of hegemons.
- Climate clubs are effective, but represent powerful interests.

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